Multiple Recovery Paths in a Lesioned Transformer

Internal Rerouting and Readout Remapping

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BlackboxNLP 2026 | Main Track | Poster #192

Motivation

A pretrained BERT encoder and a learned linear readout classify each word into one of 17 part-of-speech categories. A large immediate accuracy drop after removing a learned sparse, task-sufficient configuration establishes necessity only for that fixed routing and readout. Allowing the residual model to adapt asks a second question: can performance return through another route, and must that recovery restore the original hidden-state representation?

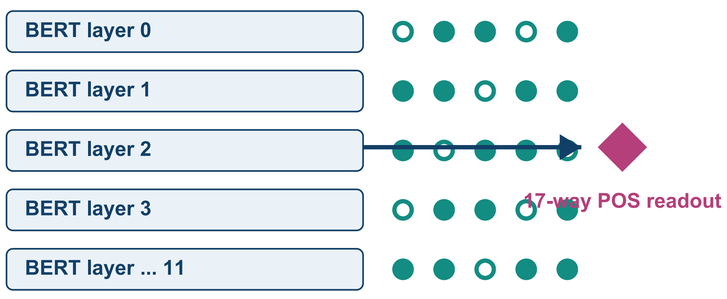
CENTRAL FINDING

High POS accuracy can return without restoring the pre-lesion encoder geometry.

Experimental framework

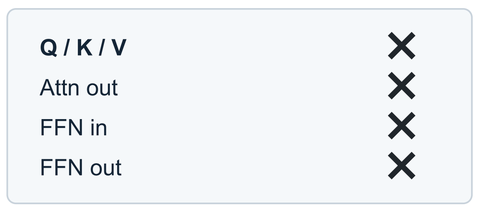
Learn a sparse task-sufficient route, hold the lesion fixed, then vary what can adapt

A Learn the sparse core



Projection-specific hard-concrete gates and a linear readout train jointly; pretrained BERT weights remain fixed.

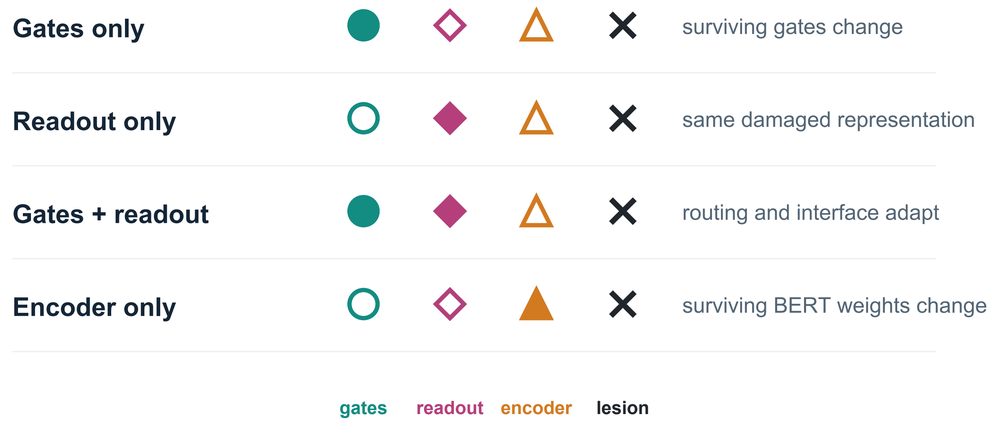
B Apply a permanent lesion



mask score = -12
gradient blocked

The selected mask entries are forced to zero and frozen. No recovery condition can reactivate a lesioned component.

C Compare recovery pathways



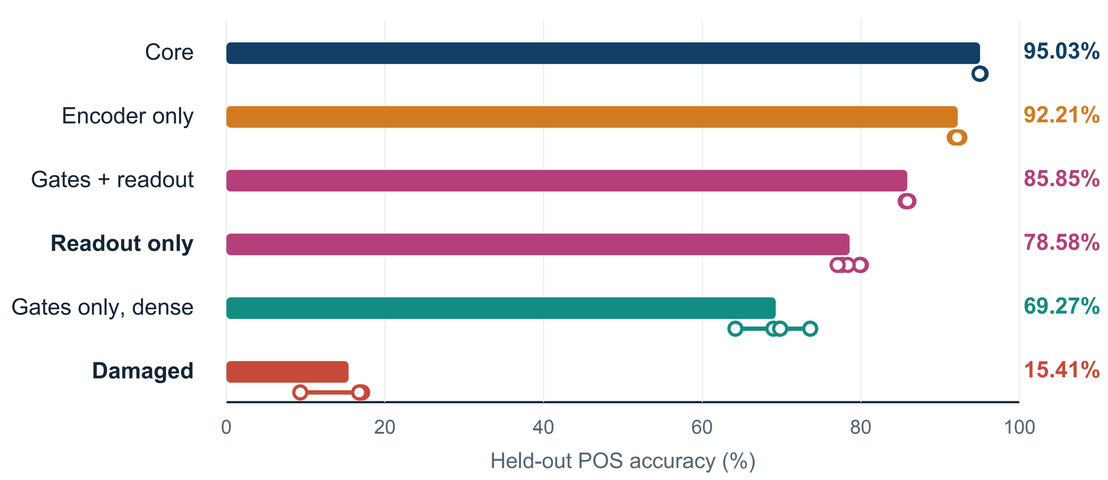
SETUP bert-base-uncased | UD English-EWT | 17 POS tags | 25,147 held-out development tokens

LESION 74 Q/K/V 64-wide input groups + 5 attention-output + 9 FFN-intermediate + 4 FFN-output gate tensors

Schematic: connections illustrate parameter status, not measured activation flow.

Recovery results

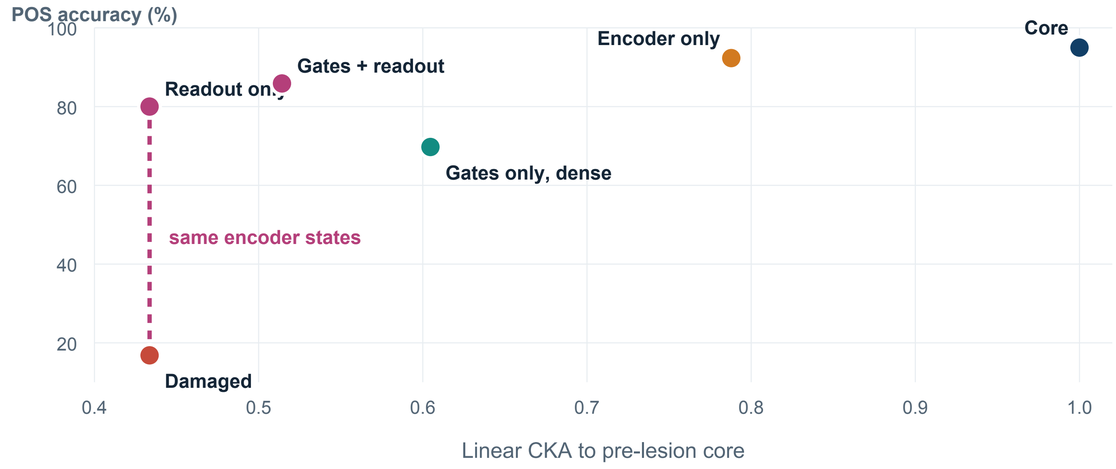
Five-seed recovery preserves the same ordering



Bars are means across five independent full-pipeline seeds; circles are individual runs and short lines show the observed range, not confidence intervals. The displayed gate-only result is the density-promoting variant. Mask-only recovery is visibly more seed-sensitive than readout, joint, or encoder recovery.

Representational analysis

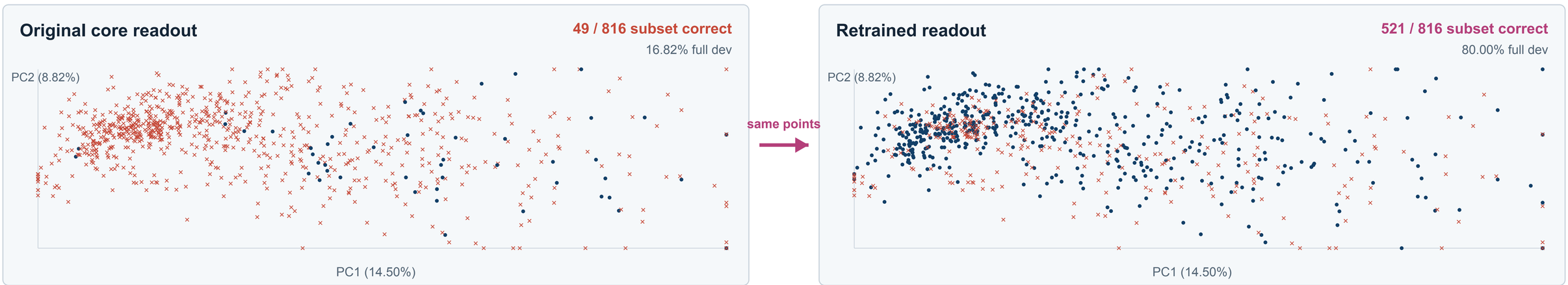
Performance and return to the core representation diverge



Linear centered kernel alignment (CKA) measures hidden-state similarity, not recovered circuitry. Primary-seed endpoints are shown because matching representations are available for that run. Readout-only stays at the damaged model's CKA by construction while accuracy rises from 16.82% to 80.00%.

Representational analysis

A new readout extracts useful POS information from the same damaged representation



Both panels contain the same POS-balanced sample of 816 token identities in one PCA basis fitted on the pre-lesion core. Coordinates remain identical; only the 17-way linear readout changes. Subset counts illustrate prediction changes; 16.82% and 80.00% are full-development accuracies over 25,147 tokens. PC1-PC2 show 23.32% of variance, not the full 768-dimensional classifier space.

Robustness and controls

Random lesions qualify what the structural intervention establishes



QKV-only controls support enrichment relative to global random groups. Full QKV+dense controls do not establish unique specificity of the exact dense target set; they validate recovery from a severe structured lesion.

Conclusions

A sparse route can be necessary in the original fixed model yet cease to be necessary once surviving parameters adapt. Readout remapping, internal rerouting, and encoder-weight retraining recover different amounts of behavior and geometry.

Scope: one BERT backbone, one supervised POS task, and a learned task readout. Recovery depends on which components may adapt; these experiments do not establish biological recovery mechanisms.

Methods: Louizos et al. (2018); Cao et al. (2021). Data: Universal Dependencies English-EWT.